

$$A = [-2; 2]$$

$$B = [0; -4]$$

$$C = [4; 0]$$

a) jsou  $A, B, C$  vrcholy  $\Delta$ ?

$$\vec{AB} = (2; -6)$$

$$\vec{AC} = (6; -2)$$

$$\left. \begin{array}{l} 2 = 6k \Rightarrow k = \frac{1}{3} \\ -6 = -2k \Rightarrow k = 3 \end{array} \right\} \frac{1}{3} \neq 3 \Rightarrow A, B, C \text{ jsou vrcholy } \Delta$$

b)  $X = [ \quad ; \quad ]$

$$\vec{AX} = 2\vec{AB} - \frac{1}{2}\vec{BC}$$

$$X - A = 2(2; -6) - \frac{1}{2}(4; 4)$$

$$X - [-2; 2] = (4; -12) - (2; 2)$$

$$X = (4; -12) - (2; 2) + [-2; 2]$$

$$X = \underline{\underline{[0; -12]}}$$

$$\vec{AB} = (2; -6)$$

$$\vec{BC} = (4; 4)$$

$$A = [-2; 2]$$

$$A = [1; -3]$$

$$B = [-5; 0]$$

$$C = [1; 3]$$

a) A, B, C vrcholy  $\Delta$

$$\vec{AB} = (-6; 3)$$

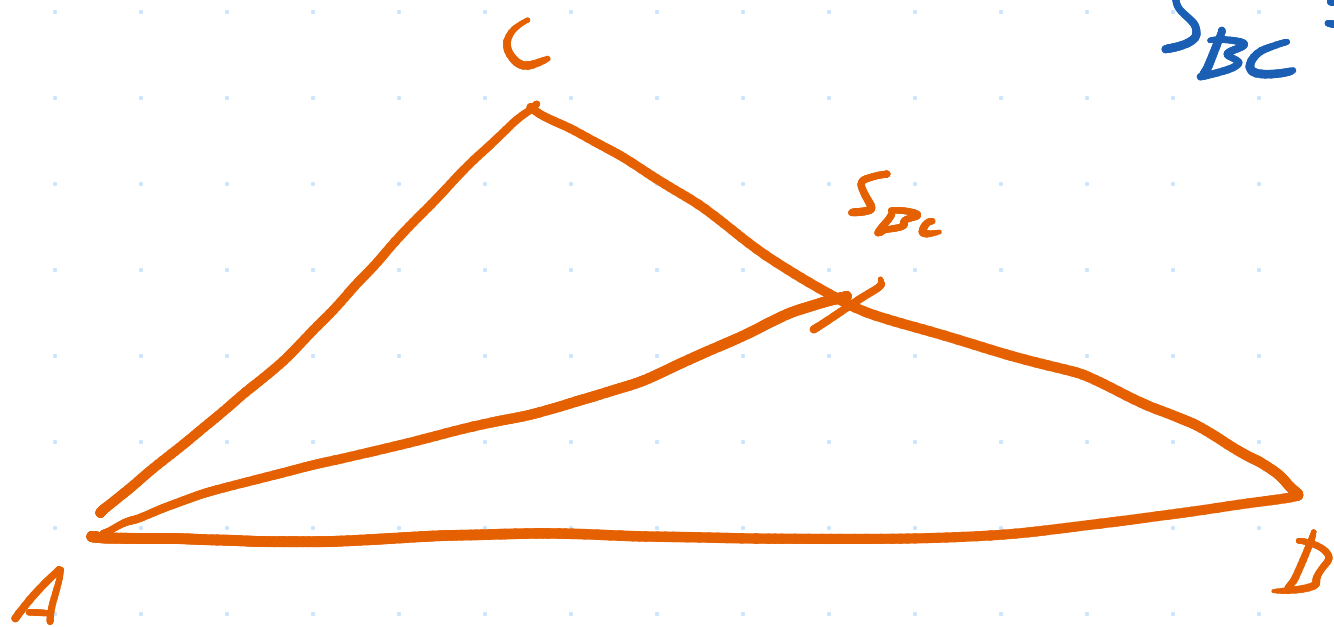
$$\vec{AC} = (0; 6)$$

$$\begin{cases} -6 = 0k \Rightarrow k = N \\ 3 = 6k \Rightarrow k = \frac{1}{2} \end{cases} N \neq \frac{1}{2}$$

$\Rightarrow A, B, C$  jsou vrcholy  $\Delta$

b) velikost  $t_a$

$$s_{BC} = [-2; 1,5]$$



$$|t_a| = |AS| = \sqrt{9 + 20,25} = \sqrt{29,25}$$

$$|AB| = \sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}$$

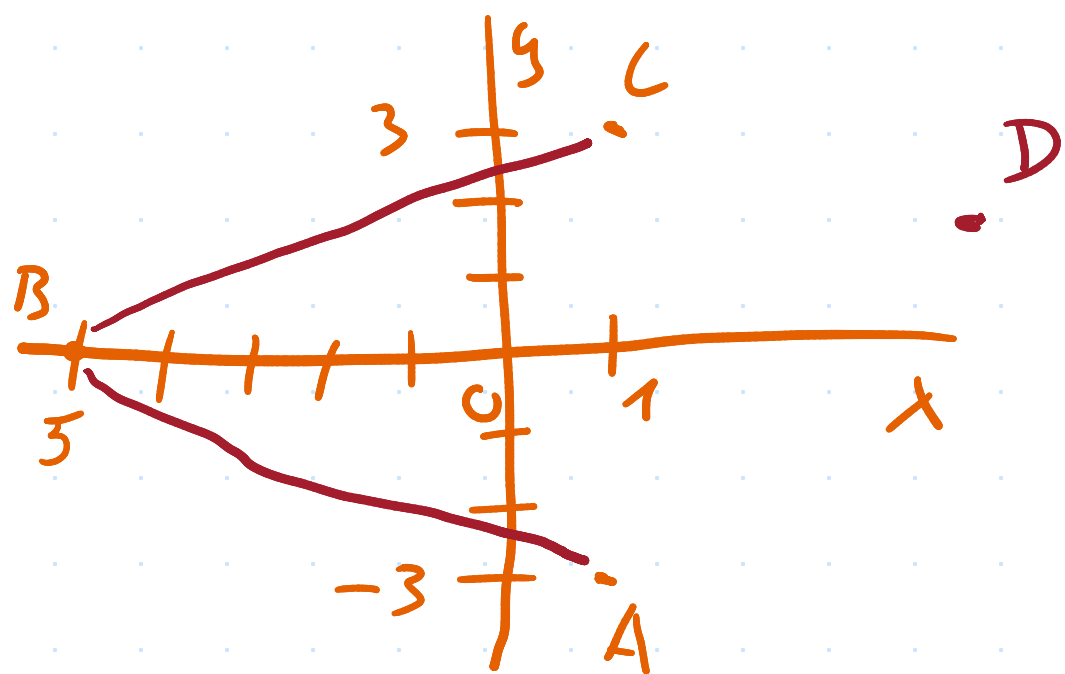
c) bod D, aby ABCD byl rovnoběžník

$$\vec{BC} = (6; 3)$$

$$D = A + \vec{BC}$$

$$D = [1; -3] + (6; 3)$$

$$D = [7; 0]$$



d) ověřte zda  $\vec{AB} \perp \vec{BC}$

$$u \vec{BC} = (6; 3)$$

$$v \vec{AB} = (-6; 3)$$

$$6 \cdot 6 + 3 \cdot 3 = 0$$

$$-27 \neq 0$$

$\vec{AB}$  je kolmý  $\vec{BC}$

$$u_x \cdot v_x + u_y \cdot v_y = 0$$

k dané přímce  $p$  sestavte parametrickou rovnoběžku  $q$ .



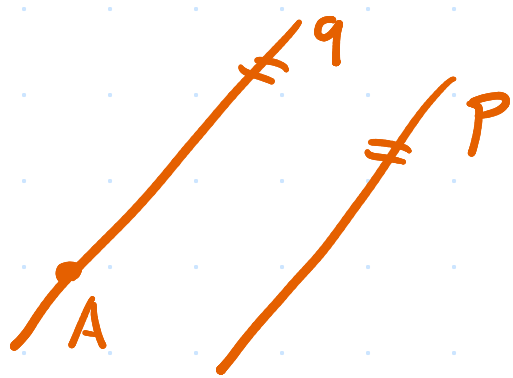
$$x = a_1 + u_1 \cdot t$$

$$y = a_2 + u_2 \cdot t$$

$$p: x - 2y + 3 = 0$$

$$A = [3; 2] \in q$$

$$\vec{p} = (1; -2) \perp (2; 1)$$



$$x = 3 + 2t$$

$$y = 2 + t$$

k přímce  $p$  napište obecnou rovnici kolmice  $q$

$$A = [2; 3] \in q$$

$$p: x = 1 + 3t$$

$$y = 2 - 2t$$

$$ax + by + c = 0$$

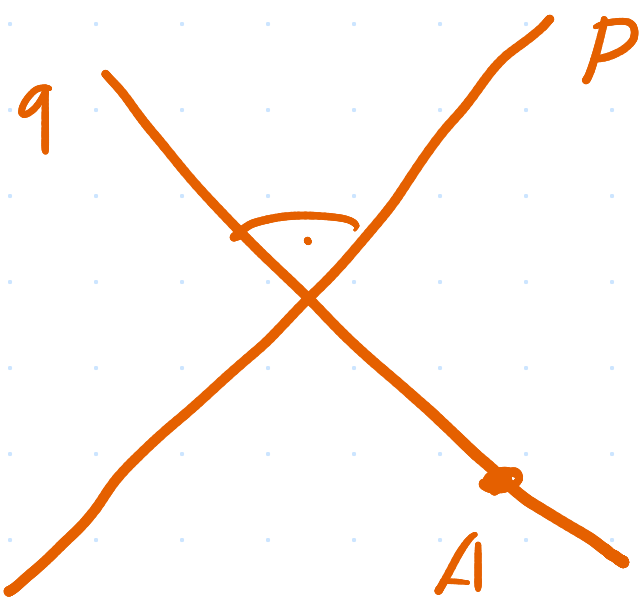
$$\vec{p} = (3; -2)$$

$$3x - 2y + c = 0$$

$$A \in q: 3 \cdot 2 - 2 \cdot 3 + c = 0$$

$$6 - 6 + c = 0$$

$$c = 0$$



$$\underline{\underline{q: 3x - 2y = 0}}$$

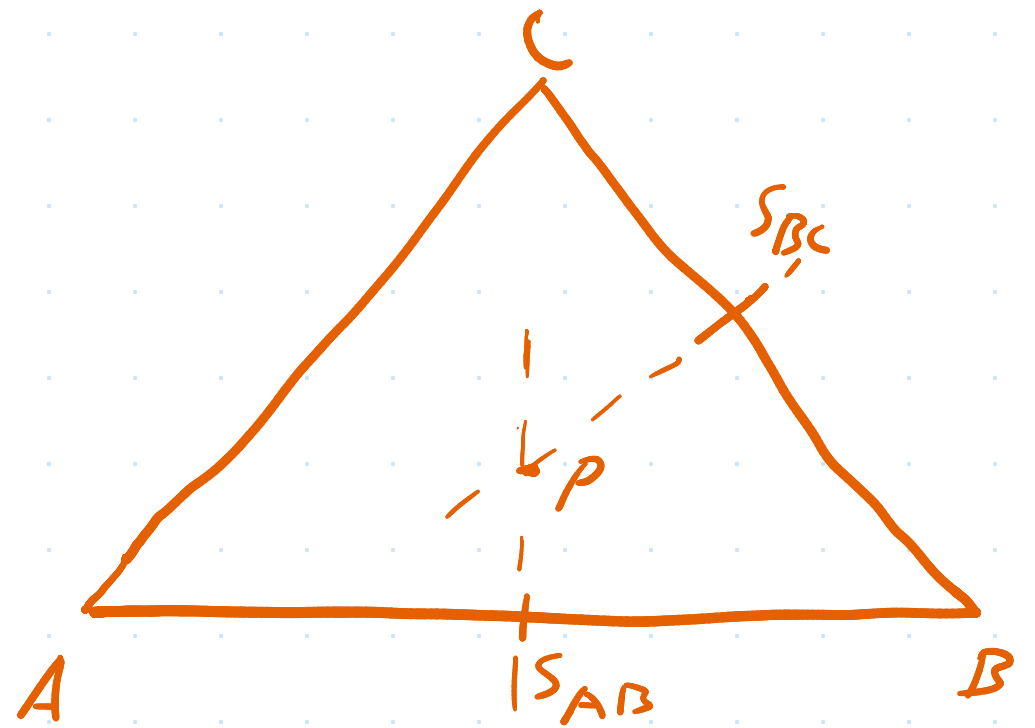


Bodů  $A, B, C$  jsou vrcholy  $\triangle ABC$ . Vypočítejte souřadnice průsečíku os stran  $AB, BC$

$$A = [-2; 4]$$

$$B = [2; 2]$$

$$C = [0; -2]$$



$$S_{AB} = [0; 3]$$

$$\vec{AB} = (4; -2)$$

$$4x - 2y + c = 0$$

$$-6 + c = 0$$

$$c = 6$$

$$4x - 2y + 6 = 0$$

$$S_{BC} = [1; 0]$$

$$\vec{BC} = (-2; -4)$$

$$-2x - 4y + c = 0$$

$$-2 + c = 0$$

$$c = 2$$

$$-2x - 4y + 2 = 0$$

$$4x - 2y + 6 = 0 \quad | \cdot (-2)$$

$$+ \begin{cases} -2x - 4y + 2 = 0 \\ \hline -8x + 4y - 12 = 0 \end{cases}$$

$$-10x - 10 = 0$$

$$-10x = 10$$

$$x = -1$$

$$-2 \cdot (-1) - 4y + 2 = 0$$

$$2 - 4y + 2 = 0$$

$$y = 1$$

$$P = [-1; 1]$$

Jedná o příčku  $p$  a bodů  $A, B$ . Vypočítejte souřadnice středu kružnice, která leží na  $p$  a kružnice prochází  $A, B$ .

$$S = [-\frac{1}{2}; \frac{5}{2}] = [-0,5; 2,5]$$

$$\vec{AB} = (3; -3)$$

$$P: x = 1 - 2t \quad A = [-2; 4]$$

$$y = -3 + t \quad B = [1; 1]$$

$$3x - 3y + c = 0$$

$$-\frac{3}{2} - \frac{15}{2} + c = 0$$

$$-\frac{18}{2} + c = 0$$

$$c = 9$$

$$p: 3x - 3y + 9 = 0$$

$$x = 1 - 2t = 1 - \frac{14}{3} = \frac{3-14}{3} = -\frac{11}{3}$$

$$y = -3 + \frac{7}{3} = -\frac{2}{3}$$

$$P = [-\frac{11}{3}; -\frac{2}{3}]$$

